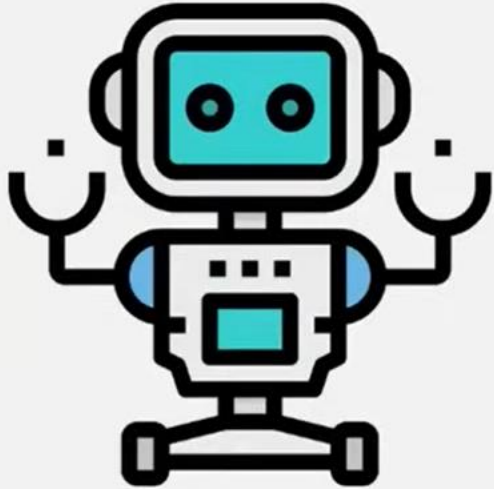
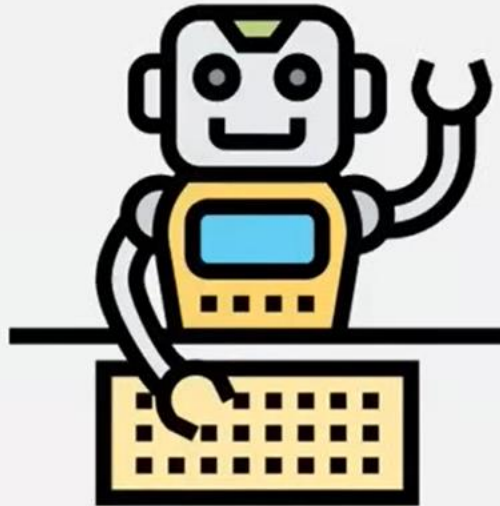


Types of Machine Learning

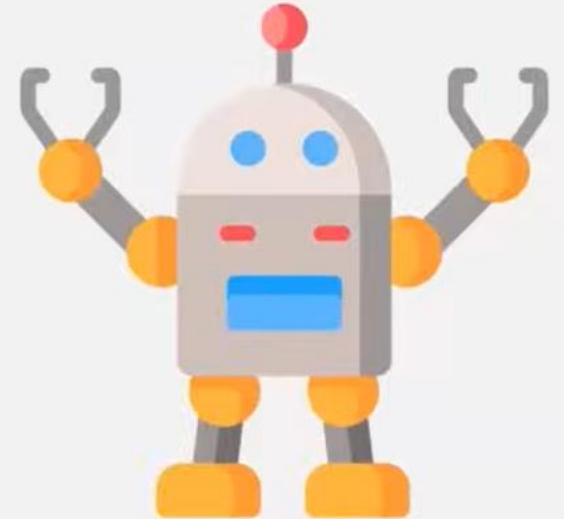
Supervised Learning



Unsupervised Learning



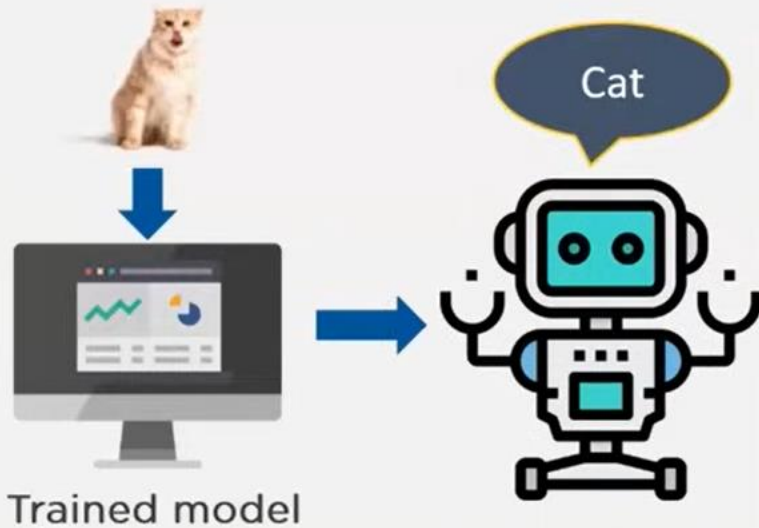
Reinforcement Learning



Definition

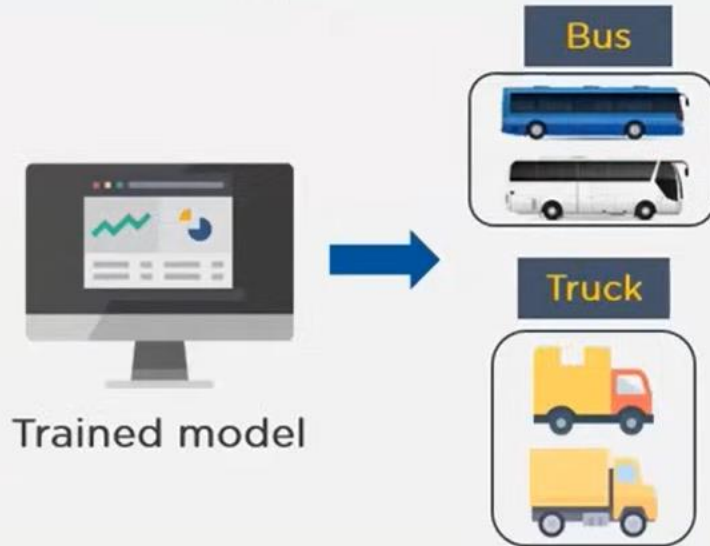
Supervised Learning

Supervised Learning is used to train machines using labeled data



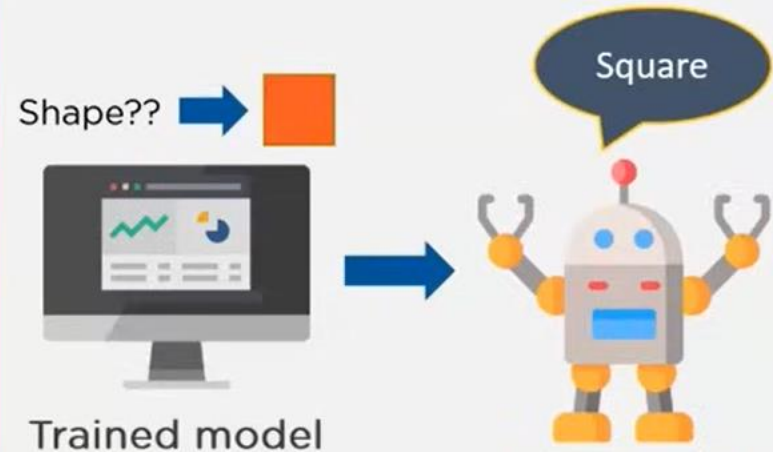
Unsupervised Learning

Unsupervised Learning uses unlabeled data to train machines



Reinforcement Learning

Reinforcement Learning uses an agent and an environment to produce actions and rewards



Algorithms

Supervised Learning

Linear Regression

Logistic Regression

Support Vector Machines

K Nearest Neighbors

Decision Tree

Unsupervised Learning

K Means Clustering

Hierarchical Clustering

DBSCAN

Principal Component
Analysis

Reinforcement Learning

Q-Learning

SARSA

Monte Carlo

Deep Q Network

Approach

Supervised Learning

Takes labeled inputs and maps it to the known outputs



Unsupervised Learning

Understands patterns and trends in the data and discovers the output



Reinforcement Learning

Follows trial and error method to arrive at the desired solution



Training

Supervised Learning

Supervised Learning techniques need external supervision to train models



Unsupervised Learning

Unsupervised Learning techniques do not need any supervision to train models



Reinforcement Learning

Reinforcement Learning techniques do not need any supervision to train models



Type of Problems

Supervised Learning

Classification and Regression



Unsupervised Learning

Clustering and Association



Reinforcement Learning

Reward based



Applications

Supervised Learning



Weather Prediction



Sales Forecast



Stock Price Analysis

Unsupervised Learning



Customer Segmentation



Churn Analysis

Reinforcement Learning

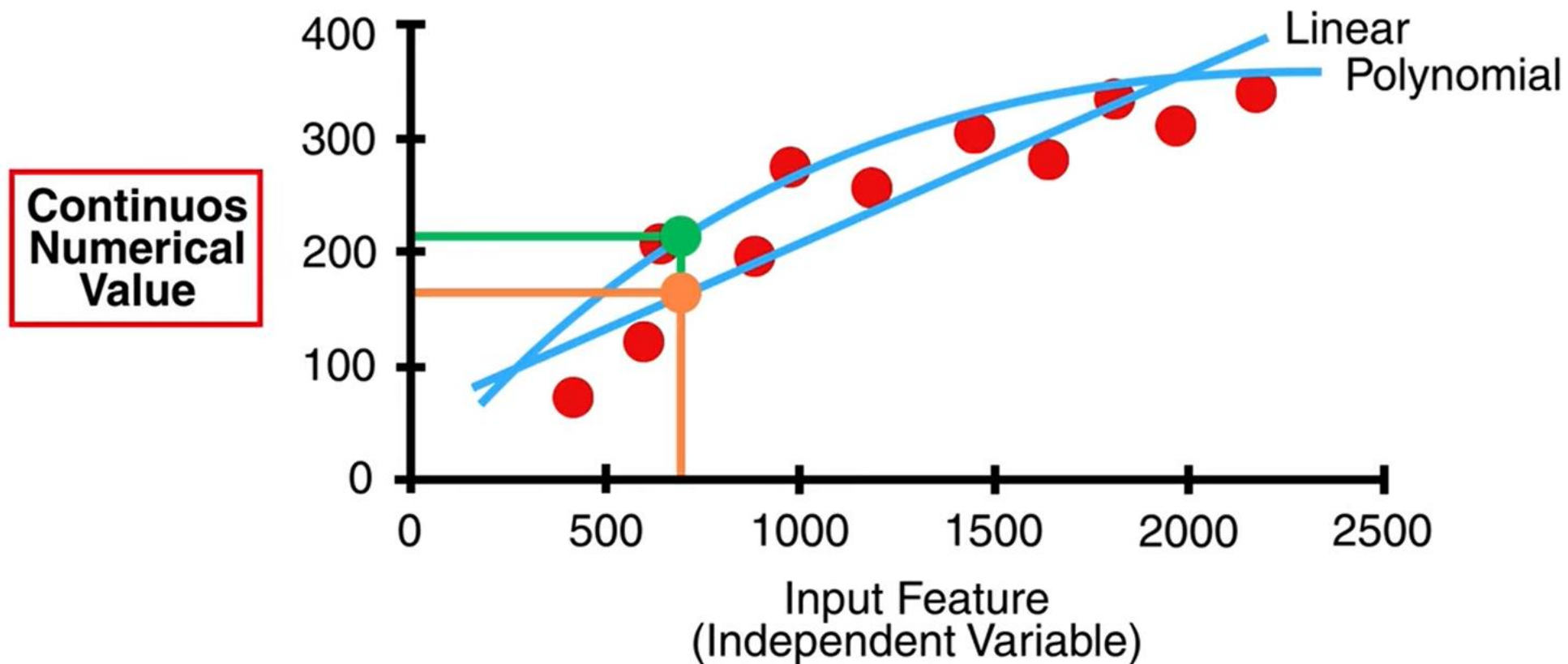


Building Games



Training Robots

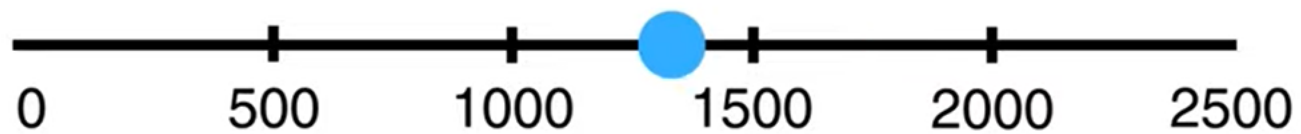
Regression



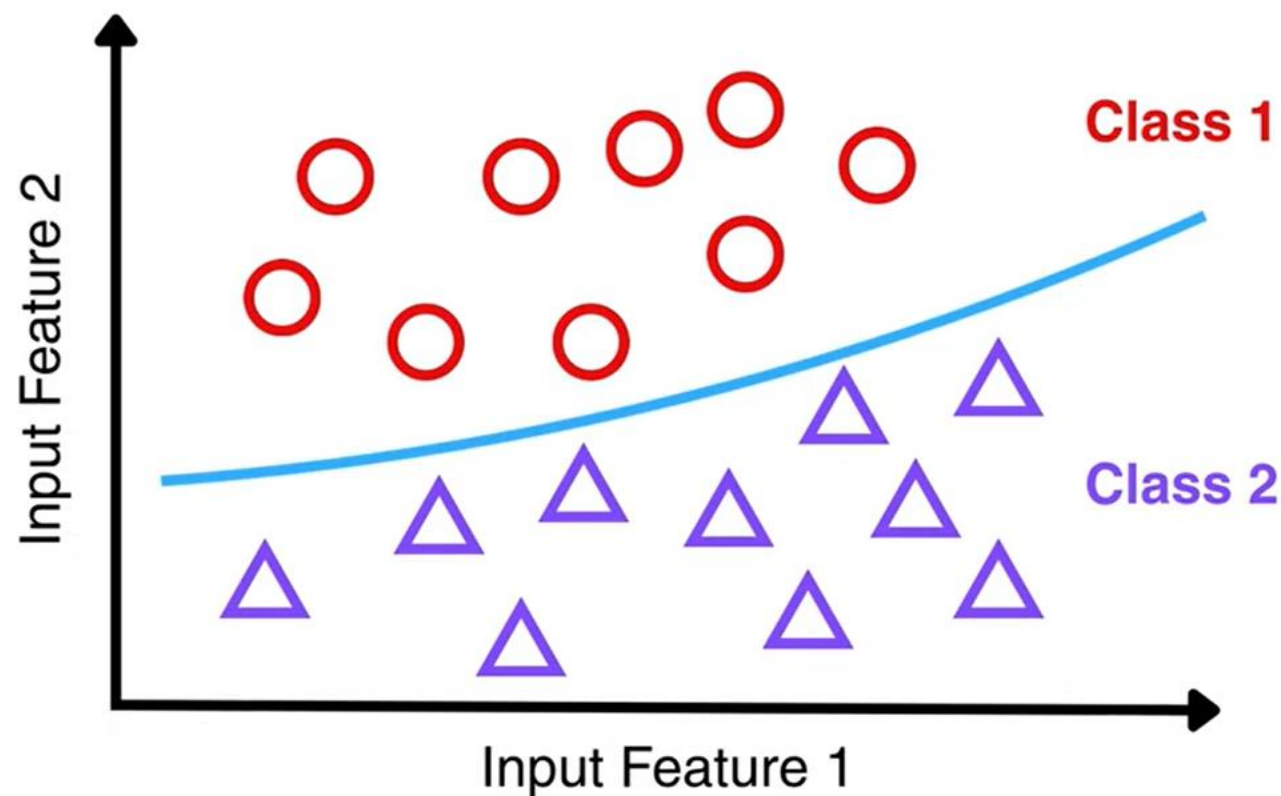
Stock Price



Business Sales

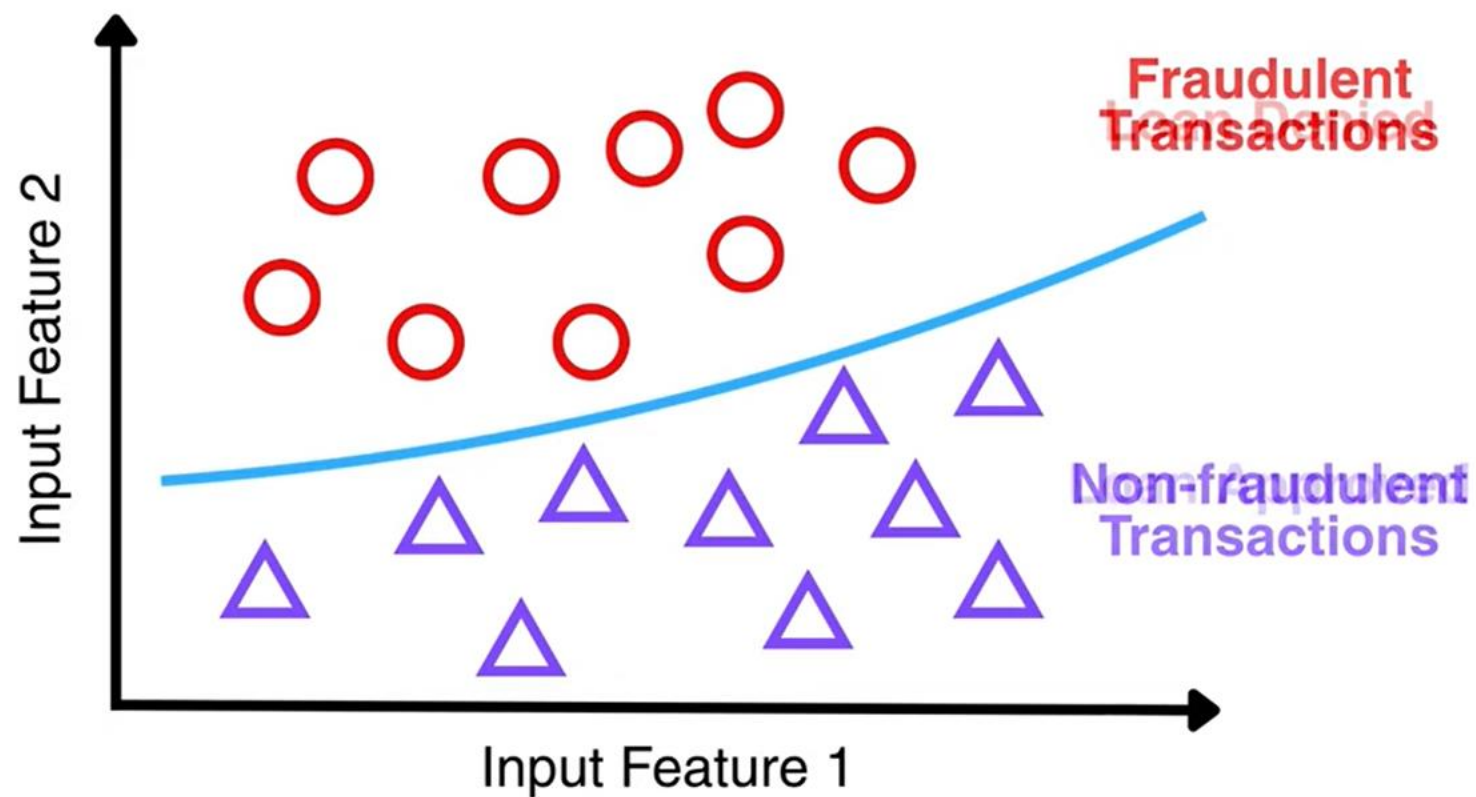


Classification

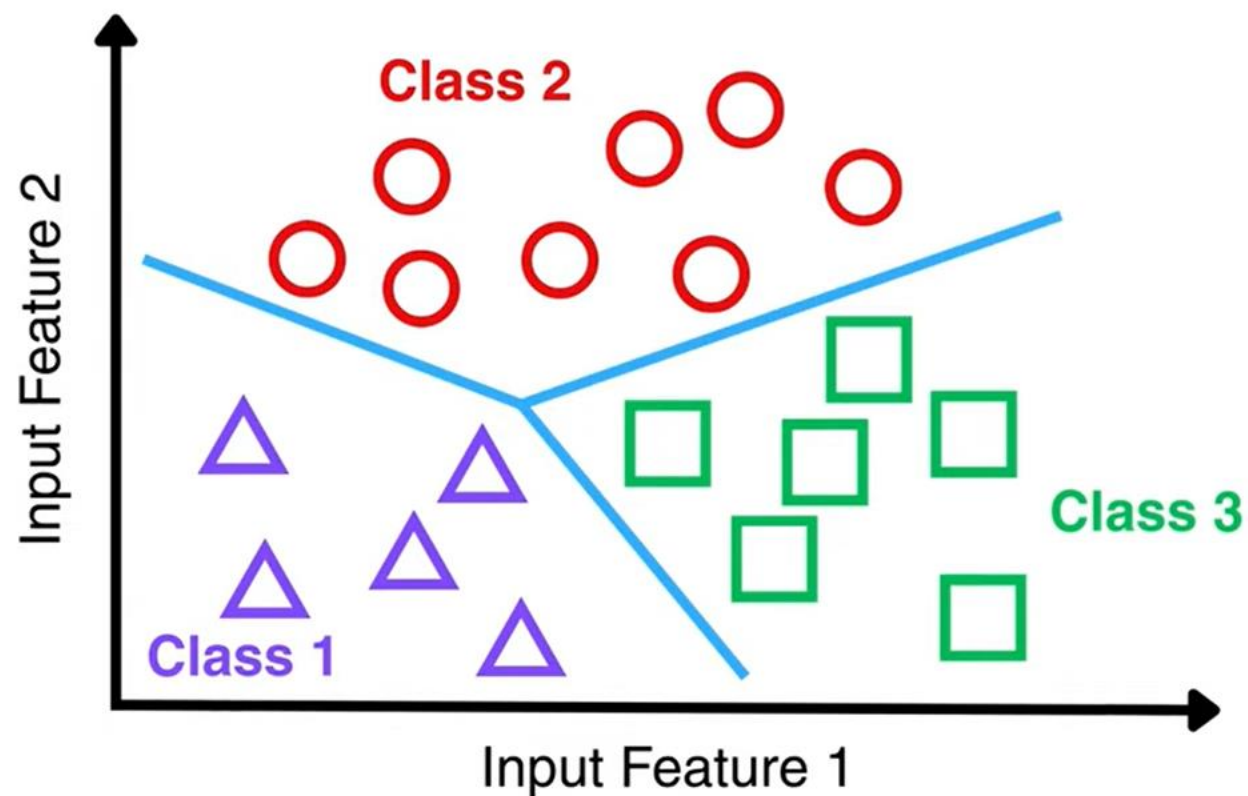


Output = Predicted **category** or **class** (from a limited set)

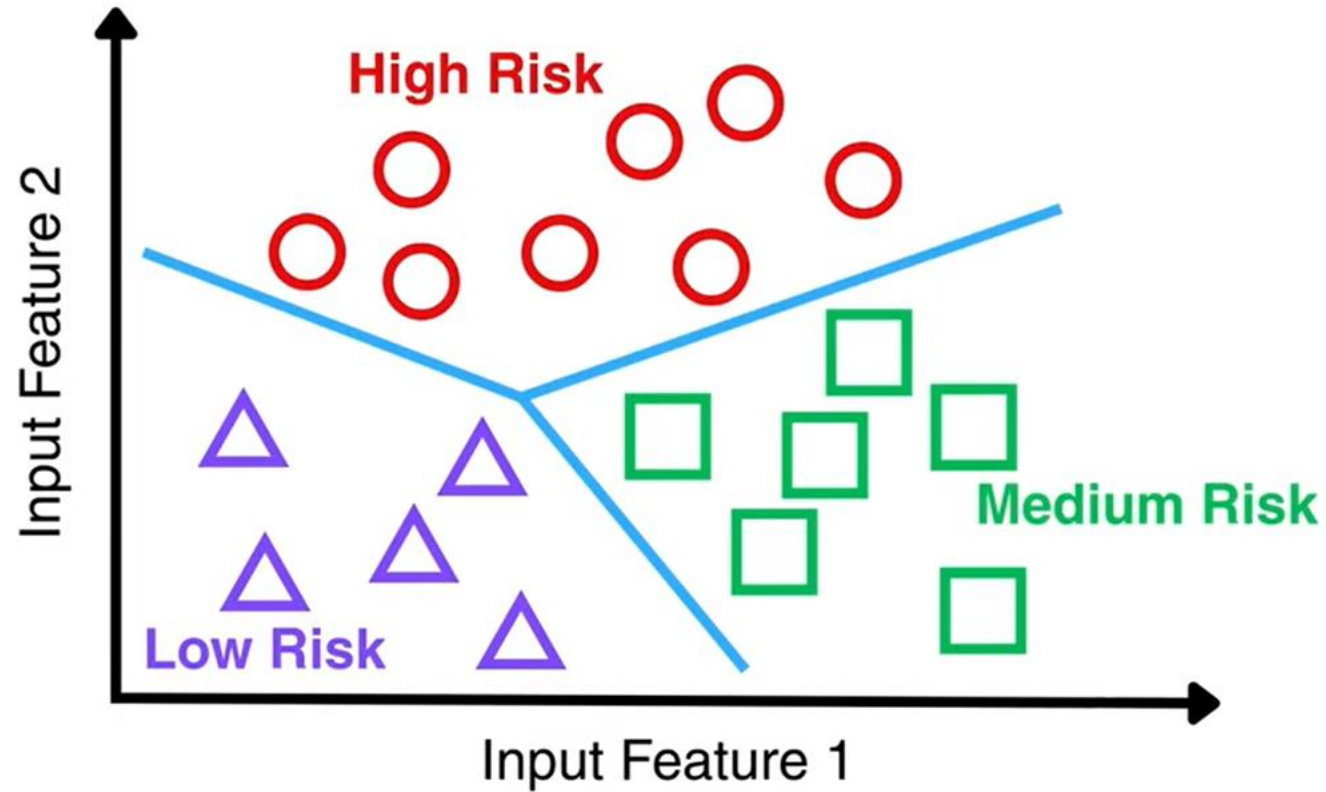
Classification



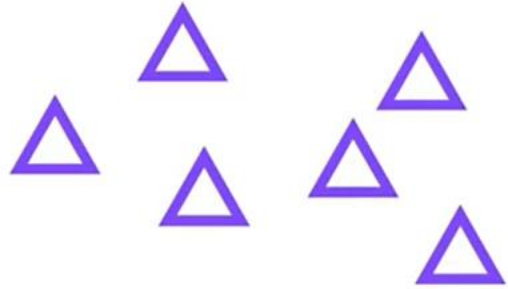
Multiclass Classification



Multiclass Classification

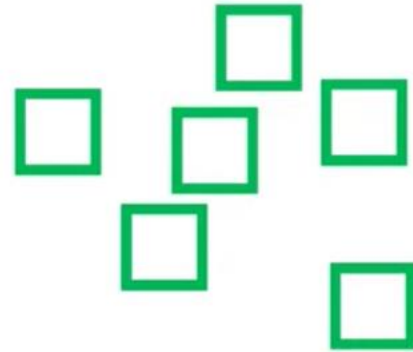


Low Risk



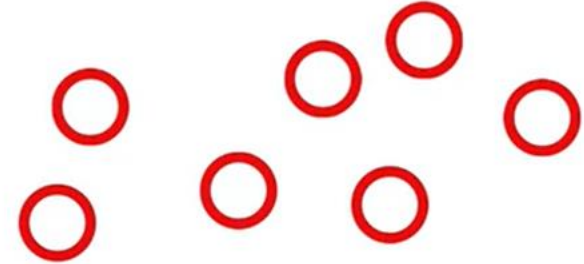
(Output: 0)

Medium Risk



(Output: 1)

High Risk



(Output: 2)

discrete classes. NOT real numerical values.